

Concept Review

Force Resistors

Why Do We Need Force Resistors?

Force sensing resistors are often used to detect changes in force, serving as pressure-sensitive buttons, or as components that monitor pressure or detect loads. These functions make them useful across a diverse range of applications such as diagnostic tools, assembly lines, robotics, and game controllers.

Force Resistors

Force resistors, or force-sensing resistors (FSRs), are variable resistors whose resistance changes when a force is applied to the surface. The resistance is typically very high with no load and decreases as more force is applied.

Force resistors contain conductive material thinly layered with a spacer separating them. These thin, flexible components can be constructed in various shapes and sizes, shown in the image below. When force is applied to the resistor, the conductive layers are pressed together, and this contact decreases resistance by allowing more electrons to flow between them.



Figure 1: Force resistors in varying shapes. Image source: <https://www.fsrtek.com/fsr>

Reading from Force Resistors

To measure from an FSR, it can be connected in a voltage divider circuit. A voltage divider circuit is shown in Figure 2 below:

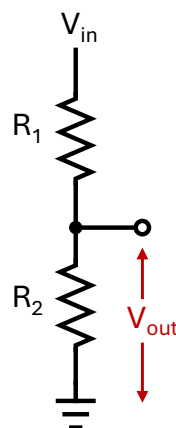


Figure 2: Voltage divider circuit

The output voltage V_{out} is a function of the input voltage V_{in} and the resistances R_1 and R_2 :

$$V_{\text{out}} = \left(\frac{R_2}{R_1 + R_2} \right) \times V_{\text{in}}$$

For a voltage divider with a force resistor as R_1 or R_2 , the output of the voltage divider changes with the resistance of the force resistor. The output voltage can be measured using the analog input pin of a microcontroller.

Datasheet and Considerations

The considerations for selecting a sensor will vary depend on specific application requirements. Performance specifications are listed in the device datasheet which should be consulted when comparing alternatives. A key specification to consider for an FSR is its **force range**, the range of force values that consistently yield measurable changes in resistance. Obtaining a reliable force estimate is unlikely beyond this range because the resistance saturates. Due to this non-linear resistance profile, calibration is not as straight-forward as sensors with linear profiles, and the resulting force estimates typically have a repeatability error of $\pm 10\%$. This is largely the reason that force sensors are not recommended for applications requiring high precision force measurement. They are widely used in wearables, user interfaces, and medical devices, offering the benefits of being a thin, flexible, and low-cost method of detecting changes in force.

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